

Mingjing Liang

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RESEARCH INTERESTS

Robot learning and autonomous decision-making, with a focus on combining model-based methods, imitation learning, and reinforcement learning to develop adaptive, robust, and data-efficient policies. Long-term interests include physically grounded learning and data-algorithm flywheels for autonomous systems.

EDUCATION

National University of Singapore

Aug 2020 - Jun 2021

Master in Mechanical Engineering

GPA: 4.5/5.0

– Relevant Modules: Deep Learning for Robotics, Neural Networks, Advanced Robotics.

National University of Singapore Research Institute (Suzhou)

Sep 2019 - Jul 2020

Joint Educational Programme in Mechanical Engineering

GPA: 4.0/4.0

– Award: Outstanding Student of Joint Educational Programme (ME-CLASS 2019)

Sichuan University

Sep 2016 - Jul 2019

Bachelor in Mechanical Engineering

GPA: 3.3/4.0

– Award: Secondary Scholarship of Sichuan University

EXPERIENCE

Autonomous Driving R&D Engineer

Sep 2021 - Present

GAC R&D Center

- Developed and optimized decision-making and planning algorithms for autonomous driving, combining rule-based and learning-based approaches
- Designed and implemented data pipelines for fleet data collection, processing, and analysis to support model iteration

PROJECTS

Large-Scale Data Pipeline for Planning

Feb 2025 - Present

- Designed and implemented a distributed data pipeline based on Ray for large-scale bag data processing
- Enabled automated data generation, filtering, and quality inspection, supporting millions of samples per batch
- Iteratively improved pipeline versions, delivering billions of high-quality data clips

Data-Driven Planning System

Jan 2023 - Jan 2025

- Architected and implemented a spatiotemporal attention-based neural planner, enabling end-to-end training and real-vehicle deployment
- Established the full data pipeline including collection standards, cleaning, augmentation, and balancing
- Integrated a trajectory optimizer as a post-processing module for real-time trajectory refinement

Bézier Curve-based Lateral Planning

Jan 2022 - Dec 2022

- Engineered a real-time path planner leveraging Bézier curves, generating smooth paths with obstacle avoidance

PUBLICATIONS AND PATENTS

Liang, Mingjing, Xun Gao, Shu Zhang, et al. (2025). “A driving trajectory planning method, electronic device, storage medium and vehicle”. Patent. URL: <https://patents.google.com/patent/CN121516035A>.

Liang, Mingjing, Xun Gao, Yulong Wang, et al. (2024). “Planning in Autonomous Driving Using Imitation Learning With Research on Data Aggregation”. In: *2024 International Symposium on Intelligent Robotics and Systems (ISoIRS)*, pp. 12–16. DOI: [10.1109/ISoIRS63136.2024.00010](https://doi.org/10.1109/ISoIRS63136.2024.00010).

Liang, Mingjing, Wenru Liu, et al. (2024). “Automatic driving model training method and device and electronic equipment”. Patent CN118520923A. URL: <https://patents.google.com/patent/CN118520923A>.

Liang, Mingjing, Huan Min, et al. (2022). “Vehicle road changing path planning method and device, storage medium and electronic equipment”. Patent CN118209121B. URL: <https://patents.google.com/patent/CN118209121B>.

SKILLS

Technical Skills	Python/C++, ROS, PyTorch, Git, Docker, Linux
Languages	IELTS (Overall 7.0)